

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 18: Wastewater Story

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	18 — Wastewater Story
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This hardest tier chains multiple plant stages and contaminant classes in one stem: it forces you to reason about what each unit physically can and cannot do, to keep aerobic aeration and anaerobic digestion (and their by-products) apart even when described by symptom rather than name, to track which of two split streams a removed material belongs to, to separate 'cleaner' from 'potable' under deceptive framings, and to place treatment events in true sequence even when listed out of order; every distractor is a precise paraphrase of a real misconception, never a giveaway. Do not write on this paper — mark answers on your answer sheet.

1. A laboratory splits one litre of raw sewage into four fractions: (i) what stays on the filter paper, (ii) the oily film floating on the filtrate, (iii) the colourless salt left when the filtrate is boiled dry, and (iv) the living things growing in a warmed sample. A student labels (iii) as 'organic matter'. Why is that label wrong?

- (A) Fraction (iii) is a dissolved inorganic impurity such as a nitrate or phosphate salt, not organic matter; the organic matter is mostly in the food and faecal solids of fractions (i) and (ii)
- (B) Fraction (iii) is wrong because the boiled residue is actually the living pathogens of fraction (iv) dried out
- (C) Fraction (iii) is wrong because dissolving in water automatically makes any substance organic, so it should be called a dissolved organic
- (D) The label is fine, because every substance recovered from sewage counts as organic matter

2. An inspector samples sewage from one shared sewer fed by homes, a dye-mill, a clinic and irrigated fields, and is asked which single source most increases the LIVING disease-causing load and which most increases the dissolved inorganic load. Which pairing is correct?

- (A) The irrigated fields add the most living pathogens, while the homes add the most dissolved inorganic salts
- (B) The clinic and homes add the most living pathogens through human and medical waste, while the dye-mill and fields add the most dissolved inorganic salts
- (C) The dye-mill adds the most living pathogens, while the clinic adds only clean rainwater
- (D) All four sources add identical amounts of every contaminant, so no source dominates any class

3. A drop of sewage is followed through a plant. Number these moments in the true order they occur: (i) a cloth rag is stopped, (ii) heavy sand drops to a chamber floor, (iii) a greasy film is skimmed from a tank surface, (iv) dissolved organic matter is eaten by oxygen-using bacteria, (v) the remaining settled solids are decomposed without air to give a flammable gas.

- (A) ii, then i, then iv, then iii, then v
- (B) i, then ii, then iii, then iv, then v
- (C) i, then ii, then iv, then iii, then v
- (D) i, then iii, then ii, then iv, then v

4. An engineer must remove from sewage, in turn, a torn sack, then road grit, then fine suspended mud, then dissolved organic matter. She wants the SINGLE unit responsible for each, in that order. Which list assigns each task to the right unit?

- (A) Grit chamber for the sack, bar screen for the grit, aeration tank for the fine mud, settling tank for the dissolved organic matter
- (B) Bar screen for the sack, settling tank for the grit, grit chamber for the fine mud, digester for the dissolved organic matter
- (C) Bar screen for the sack, grit chamber for the grit, aeration tank for the fine mud, settling tank for the dissolved organic matter
- (D) Bar screen for the sack, grit-and-sand chamber for the grit, settling tank for the fine suspended mud, aeration tank for the dissolved organic matter

5. A plant has two biological tanks. In Tank P air is bubbled continuously and the water clears of dissolved organic matter; in Tank Q the contents are sealed from air and a flammable gas collects at a dome. A trainee writes that Tank P uses anaerobic bacteria to make biogas and Tank Q uses aerobic bacteria to clean water. How many of his two claims are wrong, and why?

- (A) Only one claim is wrong, because Tank P really does make biogas though Tank Q is correctly described
- (B) Neither claim is wrong, because aeration always produces biogas and sealing always cleans water

(C) Both claims are wrong: Tank P is aerobic and cleans the water, while Tank Q is anaerobic and produces biogas

(D) Both claims are wrong, but only because both tanks actually use the very same aerobic bacteria

6. Two solids both sink to a tank floor in the plant: one is hard mineral sand, the other is mostly decaying organic matter. A worker proposes feeding BOTH to the sealed digester to make biogas. Which judgement is correct?

(A) Both should go, because the grit is the main raw material the anaerobic bacteria turn into biogas

(B) Neither should go, because the digester acts only on the floating scum, not on anything that settles

(C) Both should go, because once a material settles it automatically becomes organic sludge

(D) Only the organic sludge should go to the digester to yield biogas; the mineral grit is inert sand that produces no biogas and was deliberately removed earlier to protect equipment

7. A downstream town points out that a plant's outlet water looks clearer than its inlet water and demands the right to pipe it straight to household taps. Which response best matches the chapter's reasoning?

(A) The outlet water is by definition pure drinking water the moment it leaves the plant, so the demand is justified

(B) Treatment greatly cuts solids and organic load so the water looks far cleaner, but it is not guaranteed free of all germs and so is still not certified fit for drinking

(C) The outlet water is actually dirtier than the inlet, so piping it to taps would poison the town

(D) The clearness proves the water now contains more dissolved nutrients, which makes it the safest drinking source

8. During a sudden cloudburst, a town's combined sewers carry an unusually heavy slug of washed-in road sand AND a surge of torn plastic and branches. An operator can only inspect two units first. Which two, matched to the two loads, should he check?

(A) The aeration tank for the plastics and the sludge digester for the road sand

(B) The settling tank for the plastics and the aeration tank for the road sand

(C) The sludge digester for the plastics and the bar screen for the road sand

(D) The bar screen for the plastics and branches, and the grit-and-sand chamber for the road sand

9. A plant's grit-and-sand chamber is bypassed by accident, so abrasive sand flows on into the rest of the works. Trace the chain of trouble the chapter's logic predicts as the sand travels forward.

- (A) The sand is the chief food of the aerobic bacteria, so cleaning speeds up and the effluent becomes drinkable
- (B) The sand dissolves into nitrate nutrients that the bacteria convert into extra biogas
- (C) The sand wears out pumps and moving parts and then settles into and contaminates the organic sludge, which lowers the quality of the sludge and the biogas/manure made from it
- (D) The sand is caught by the bar screen further downstream, so no harm reaches any later unit

10. A worker stares into a tank and sees a dense layer being scraped along the floor toward an outlet, while a separate arm sweeps a greasy film off the surface, and notices NO air bubbling. Which tank is it, and what are the bottom and top materials called?

- (A) It is the aeration tank; the bottom layer is activated sludge and the surface film is biogas
- (B) It is the settling tank; the bottom layer is scum and the surface film is sludge
- (C) It is the sludge digester; the bottom layer is grit and the surface film is manure
- (D) It is the settling tank; the bottom layer scraped along is sludge and the surface film is scum

11. A simplified plant is built with the aeration tank LEFT OUT: screened, de-gritted, settled and clarified water is sent straight to chlorination and then to the river. Which problem follows MOST directly from omitting only the aeration step?

- (A) Large plastic bags now pour uncaught into the river
- (B) Heavy grit and sand are no longer removed from the water
- (C) Dissolved organic matter stays in the water because the aerobic bacteria that would have eaten it never got the chance
- (D) The bar screen can no longer catch any debris

12. A claim circulates: 'Chlorination at the end of treatment is what removes the dissolved nitrates and phosphates from the water.' Why is this wrong?

- (A) It is wrong because chlorination's real job is to settle out heavy grit and sand
- (B) It is wrong because chlorination only produces biogas from leftover organic matter
- (C) It is correct, because chlorine dissolves and washes away all dissolved salts
- (D) Chlorination is added to kill harmful microbes the earlier steps missed; it does not remove the dissolved nitrate and phosphate salts, which are a separate dissolved-inorganic load

13. Inside the plant the flow splits after the settling tank into a 'solids' stream and a 'liquid' stream. A student must say which stream goes to the air-free digester and which to the aerated tank, and what each yields. Which mapping is fully correct?

- (A) The settled sludge (solids) goes to the air-free digester and yields biogas and digested sludge; the clarified water (liquid) goes to the aerated tank and is cleaned of dissolved organics
- (B) The settled sludge goes to the aerated tank and yields biogas; the clarified water goes to the air-free digester and is cleaned
- (C) Both the sludge and the clarified water go together into the one air-free digester to make biogas
- (D) The clarified water goes to the digester to make manure; the sludge is poured straight back into the inlet

14. A village biogas plant fed with cattle dung and toilet waste produces a flammable gas. A teacher asks which plant unit shares the SAME air condition and the SAME kind of by-product. Which answer is right?

- (A) The sealed sludge digester, because both work without air (anaerobic) and both yield biogas
- (B) The aeration tank, because both work with plenty of air and both yield biogas
- (C) The grit chamber, because both rely on sunlight and both yield manure
- (D) The bar screen, because both rely on coarse bars and both yield a flammable gas

15. Three impurities are pulled from one sample: human faeces, dissolved nitrate, and intestinal bacteria. A chemist must drop each into exactly one of three boxes labelled 'organic matter', 'dissolved inorganic' and 'living pathogen'. Which assignment is correct?

- (A) Faeces into dissolved inorganic, dissolved nitrate into living pathogen, intestinal bacteria into organic matter
- (B) Faeces into living pathogen, dissolved nitrate into organic matter, intestinal bacteria into dissolved inorganic
- (C) All three into the single box of dissolved inorganic impurities
- (D) Faeces into organic matter, dissolved nitrate into dissolved inorganic, intestinal bacteria into living pathogen

16. A plant operator notices the treated water is leaving with an unusually HIGH dissolved organic content even though the bar screen, grit chamber and settling tank all look normal and the sludge is being digested. Which single failure best explains this, given what each unit can do?

- (A) The bar screen has failed, because the bars are what strain dissolved organic matter from the water
- (B) The aeration tank's bacteria are not working properly, because that is the only stage that removes dissolved organic matter
- (C) The grit chamber has failed, because settling sand is what carries away dissolved organic matter
- (D) The settling tank has failed, because it is the only place dissolved organic matter is removed

17. A hill hamlet with no underground sewer wants each house to handle its own toilet waste cheaply and on-site, and asks what must be done as the years pass. Which choice fits, and what is the recurring task?

- (A) A full municipal treatment plant for each house, which never needs emptying because it consumes all its own sludge
- (B) A bar screen under each toilet, whose collected debris turns itself into clean drinking water over time
- (C) A grit-and-sand chamber for each house, whose contents automatically become kitchen cooking gas
- (D) A septic tank, whose accumulated settled sludge must be periodically pumped out and safely disposed of or treated

18. A claim reads: 'The biogas drawn from the digester dome and the scum skimmed off the settling tank are really the same substance, just collected at different points.' Identify the precise reason this is false.

- (A) It is false only because biogas is the heavy sludge at the bottom while scum is a gas
- (B) Biogas is a flammable gas released by anaerobic decomposition of sludge, whereas scum is a floating oily/greasy solid film skimmed from the surface earlier — different states, sources and uses
- (C) It is true, because both are the floating oily layer that rises to the top of a tank
- (D) It is true, because both are flammable gases collected from sealed domes

19. A factory secretly pipes strong acids and toxic salts into the public sewer. Beyond polluting the river, the chapter warns this hurts the treatment PROCESS itself. Through which mechanism, and which stage is most directly crippled?

- (A) The chemicals make the bar screen catch debris faster, crippling only the screening stage
- (B) The chemicals dissolve the grit, crippling only the grit chamber
- (C) The chemicals turn the sewage instantly into safe manure, so no stage is harmed at all
- (D) The toxic chemicals can kill the helpful bacteria, most directly crippling the biological aeration and digestion stages that depend on living microbes

20. At one corner of a plant a worker sees a sealed dome with a flammable gas being drawn off and no air being pumped in; nearby a separate open tank shows vigorous bubbling and clearing water. Which statement correctly names BOTH units and their by-products?

- (A) The sealed dome is the aeration tank giving oxygen; the bubbling tank is the digester giving biogas
- (B) The sealed dome is a settling tank giving scum; the bubbling tank is a grit chamber giving sand

- (C) The sealed dome is the anaerobic digester giving biogas; the bubbling tank is the aerobic aeration tank giving cleaned water and activated sludge
- (D) Both units are digesters and both give off biogas

21. A student argues that pouring solid food scraps and tea leaves down the sink is harmless because 'they are natural, organic and so the bacteria will simply eat them.' Where does this reasoning break down?

- (A) It breaks down because the scraps dissolve at once and improve the water quality
- (B) It breaks down because aerobic bacteria living inside the home taps eat them immediately
- (C) Being solid, the scraps build up and choke the pipes before they ever reach the bacteria, obstructing the flow of sewage regardless of being organic
- (D) It does not break down at all; pouring food scraps down the sink is genuinely harmless

22. A town suffers repeated cholera and typhoid outbreaks. Investigators find leaking sewers seeping into shallow drinking-water wells. The chapter would identify the root cause and the broad remedy as which pair?

- (A) The root cause is sewage mixing with the water people drink; the remedy is better sanitation that keeps sewage away from drinking-water sources
- (B) The root cause is too much biogas produced in homes; the remedy is to ban household cooking gas
- (C) The root cause is treated water being recharged into the ground; the remedy is to stop all groundwater recharge
- (D) The root cause is bar screens catching too many plastics; the remedy is to remove the screens

23. Most of the bacterial mass that settles at the bottom of the aeration tank after feeding on organic matter is given a name, and a large part of it is given a deliberate fate. Which statement captures both correctly?

- (A) It is called grit, and it is sent to the digester as the raw material for biogas
- (B) It is called scum, and it is skimmed off and burnt directly as cooking fuel
- (C) It is called raw sewage, and it is poured straight back into the plant inlet untreated
- (D) It is called activated sludge, and much of it is pumped back into the aeration tank to keep the bacterial cleaning going

24. A vermi-processing toilet and a plain flush toilet emptying to a river are compared. Which single contrast captures why the vermi toilet is called environment-friendly?

- (A) The vermi toilet uses earthworms to convert the human waste into useful manure on site, instead of releasing untreated waste into a river
- (B) The vermi toilet flushes the waste faster and further into the nearest river, so it spreads out the pollution

(C) The vermi toilet burns the waste with no organisms and pipes the resulting biogas to the kitchen

(D) The vermi toilet pumps a steady stream of air to grow aerobic bacteria exactly like a mini treatment plant

25. A simplified plant is built that KEEPS aeration but OMITTS the settling (sedimentation) tank, sending grit-removed water straight into the aeration tank. Which problem follows most directly?

(A) Fine suspended solids that the grit chamber missed are never settled out, so they enter the aeration tank and burden it instead of being removed first as sludge

(B) Dissolved organic matter can no longer be removed anywhere in the plant

(C) Large plastic bags now flow uncaught into the river

(D) The grit and sand are no longer removed from the water

26. A worker compares the solids at the BOTTOM of the settling tank with the solids at the bottom of the aeration tank. Which statement distinguishes them correctly?

(A) Both bottoms hold the same heavy mineral grit removed earlier

(B) Both bottoms hold the same flammable biogas collected from below

(C) The settling tank's bottom holds raw settled sludge of mixed organic solids, while the aeration tank's bottom holds activated sludge made of the bacteria that cleaned the water

(D) The settling tank's bottom holds biogas and the aeration tank's bottom holds floating scum

27. Treated effluent that is not certified for drinking is pumped into the ground rather than discarded. A student offers four reasons; pick the one that matches the chapter.

(A) It is already pure drinking water, so recharging it merely stores spare drinking water underground

(B) Though not potable, the cleaned water is safe enough to soak into the soil and help replenish the groundwater supply

(C) It is dirtier than raw sewage and must be hidden underground where no one can find it

(D) Soaking it into the ground converts the effluent into collectable biogas

28. A claim states: 'After sewage passes the bar screen AND the grit chamber, every solid impurity has been taken out of the water.' Why is this judgement wrong?

(A) Only bulky objects and heavy mineral grit are gone; finer suspended solids, the floating scum, and the dissolved and microscopic impurities all remain to be dealt with later

(B) It is wrong only because the bar screen actually adds new solids back into the water

(C) It is correct, because between them the screen and the grit chamber remove every solid completely

(D) It is wrong only because the grit chamber removes the dissolved salts but not the bulky objects

29. Rank what physically LEAVES the main water stream from earliest to latest: (i) settled sludge drawn from the settling-tank floor, (ii) heavy grit, (iii) large floating debris, (iv) dissolved organic matter consumed by bacteria.

(A) i, then ii, then iii, then iv

(B) ii, then iii, then i, then iv

(C) iv, then i, then ii, then iii

(D) iii, then ii, then i, then iv

30. Untreated sewage rich in dissolved nitrates and phosphates is released into a still pond, and weeks later the pond is choked with overgrown plants and the fish are dying. Which explanation matches the chapter's account of these dissolved 'nutrients'?

(A) The dissolved nitrate and phosphate over-nourish the water plants, whose excessive growth upsets the pond's natural balance and harms the other life in it

(B) The dissolved nutrients made the pond water immediately fit for drinking, which killed the fish by being too pure

(C) The dissolved nutrients sank and hardened into harmless construction sand on the pond floor

(D) The dissolved nutrients were converted by the pond into collectable biogas, which suffocated the plants

31. A campsite far from any sewer or treatment plant fits its toilets with a chemical-dosing system. What is the purpose of the chemicals there, and why is a chemical toilet chosen over piping to an aeration tank?

(A) The chemicals turn the waste straight into safe drinking water for the campers

(B) The chemicals break down the waste and cut its odour and germs on the spot, which suits a place with no sewer or plant to send the waste to

(C) The chemicals grow aerobic bacteria that require a constant powered air pump the campsite must run

(D) The chemicals collect flammable biogas that is piped to the camp kitchen for cooking

32. An auditor lists the genuinely useful, sellable PRODUCTS recovered across the whole treatment process. Which pair is truly useful, and from which processes do they come?

(A) Plastic bags from the bar screen as a building material, and grit from the chamber as a fertiliser

(B) Biogas as a fuel, from anaerobic digestion of sludge, and dried digested sludge as manure, from the same digester's residue

(C) Floating scum as a clean cooking fuel, and grit as a soil nutrient

(D) Raw untreated sludge tipped into the river as fish food, and chlorine as a drink additive

33. At the sink a family must decide which single act is **LEAST** likely to either clog the pipes or poison the treatment bacteria. Which choice is the safe one, and what disqualifies the others?

- (A) Tipping out leftover paint thinner, since solvents only help the bacteria work faster
- (B) Pouring hot used frying oil, since heat keeps it from ever coating the pipes
- (C) Pouring plain water used to rinse a clean glass, since it adds no solids to clog and no toxins to harm the bacteria
- (D) Flushing a handful of expired tablets, since medicines are eaten harmlessly by the bacteria

34. A municipal crew that sweeps streets and clears blocked public drains and toilets is said to protect public health. By the chapter's reasoning, what is the chain from this routine work to fewer diseases?

- (A) Sweeping and clearing deliberately raises the number of microbes reaching household taps, which trains people's immunity
- (B) The crew adds extra chemicals into the public's household sinks, which sterilises the drinking supply
- (C) The work speeds up the burning of biogas in people's kitchens, which warms the water and kills germs
- (D) Clearing drains stops them overflowing and keeps disease-carrying waste away from where people live and gather, which cuts the chance of that waste contaminating water and spreading illness

35. A teacher writes four 'facts' on the board and asks which is the **ONLY** fully correct one about the treatment plant.

- (A) The grit chamber removes dissolved nitrates while the digester removes large plastic bags
- (B) The settling tank produces biogas while the aeration tank produces grit
- (C) The bar screen removes impurities purely physically with no bacteria, while the aeration tank removes dissolved organics biologically with aerobic bacteria
- (D) The bar screen kills the disease microbes while the chlorination step catches the large debris

36. A student insists, 'Because the digester is sealed and air-tight, it must be growing aerobic bacteria inside, since sealed places trap air.' Untangle the error.

- (A) Sealing keeps air out so that anaerobic bacteria, which work without oxygen, can decompose the sludge and release biogas
- (B) Sealing traps fresh air inside, so aerobic bacteria do thrive and clean the water there
- (C) Sealing makes no difference to the bacteria, because both tanks use identical bacteria anyway
- (D) Sealing lets sunlight in to dry the sludge, so the digester grows no bacteria at all

37. Open defecation occurs in fields just uphill of a stream that a downstream village drinks from. The chapter's reasoning links this to disease through which chain?

- (A) The exposed waste quickly turns into harmless biogas in the open air, so no disease results
- (B) The waste increases the village's supply of clean drinking water by adding to the stream
- (C) Rain washes the exposed waste and its germs downhill into the stream, contaminating the drinking water and spreading waterborne disease
- (D) The waste feeds helpful aerobic bacteria that live inside the village's taps and keep them clean

38. A plant's aeration blower fails for two days while everything else runs. Predict, in order, what happens to (a) the dissolved-organic cleaning and (b) the activated-sludge bacteria.

- (A) Freed from the disturbance of bubbling, the aerobic bacteria thrive even more and clean the water better
- (B) The tank switches to making biogas in place of clean water while the bacteria multiply faster
- (C) Starved of oxygen, the aerobic bacteria can no longer thrive, so the dissolved organic matter is broken down poorly and the activated-sludge population declines
- (D) Gravity alone now filters out the dissolved organic matter, so the bacteria become unnecessary

39. A worker must point to the ONE step that removes impurities by purely physical action with no help from living organisms, distinguishing it from the biological steps. Which is it, and why are the others disqualified?

- (A) The bar screen catching bulky objects, because it is a passive sieve; the aeration tank and the sludge digester both rely on living bacteria
- (B) The aeration tank consuming dissolved organic matter, because its bubbling is purely mechanical
- (C) The sludge digester decomposing sludge, because sealing it removes any role for organisms
- (D) The recycled activated sludge cleaning the water, because recycling is a purely mechanical pump action

40. A claim: 'Activated sludge and the dried digested sludge sold as manure are the same material at different times.' What is the precise correction?

- (A) They are the same, because both are simply dried bottom solids from the same tank
- (B) The correction is that activated sludge is a flammable gas while manure is the bottom grit

(C) Activated sludge is the living mass of aerobic bacteria settling in the aeration tank, whereas manure is the dried residue left after anaerobic digestion of sludge in the digester — different stages, processes and states

(D) They are the same, because both are the floating scum skimmed at the surface

41. Two materials both float to the top of tanks at a plant: a greasy film on the settling tank and the gas drawn from the digester dome. A student says 'both float, so both are scum.'

Why is this wrong, and what is each correctly?

(A) Both are correctly scum, since anything that rises in a tank is by definition scum

(B) Only the greasy film is scum; the digester output is biogas, a flammable gas that rises and is collected, not a floating oily skimmed layer

(C) Neither is scum; the greasy film is biogas and the dome gas is sludge

(D) Both are biogas, since both are collected from the tops of tanks

42. A plant operator finds the treated water leaving with far MORE bacteria than usual, although the aeration tank, settling tank and digester are all working and the water is otherwise clear. Which single failure best explains the spike in microbes specifically?

(A) The grit chamber has failed, because removing sand is what kills the microbes

(B) The final chlorination/disinfection step has failed, because that is the stage meant to kill the harmful microbes the other steps leave behind

(C) The bar screen has failed, because straining large debris is what removes the microbes

(D) The aeration tank has failed, because aeration is the only step that kills microbes

43. A student claims a single statement is true; pick the genuinely correct one about where each contaminant class is dealt with.

(A) Bulky solids, dissolved organic matter and microbes are all removed together by the bar screen

(B) Bulky and suspended solids are removed by screening and settling, dissolved organic matter is consumed by aerobic bacteria in aeration, and remaining harmful microbes are tackled by disinfection

(C) Dissolved organic matter is screened out and the bulky solids are eaten by the aeration bacteria

(D) Microbes are settled out as sludge while the dissolved salts are caught by the bar screen

44. A claim about ordering: 'It does not matter whether grit removal comes before or after aeration, because both just take stuff out of water.' Why does the chapter insist grit removal comes BEFORE aeration?

(A) Because the aeration bacteria need the sand as their main food, the grit must arrive first to feed them

(B) Because the aeration tank removes grit more thoroughly than the grit chamber, so grit waits until then

- (C) If grit reached the aeration tank, the abrasive sand would wear out the aeration equipment and could smother the working bacteria, so it must be removed first
- (D) Because grit must be aerated before it is heavy enough to settle out

45. A village installs a community biogas plant fed with dung and toilet waste, and notices the gas output collapses after a neighbour pours strong bleach and acid into the feed. What is the most likely reason, and what does it reveal about the gas-making bacteria?

- (A) The chemicals simply diluted the dung, showing biogas is a purely chemical reaction needing no organisms
- (B) The harsh chemicals killed the anaerobic bacteria that make the biogas, showing the gas depends on living microbes that toxins can destroy
- (C) The chemicals added oxygen, which made the anaerobic bacteria work even faster and overflow the gas
- (D) The chemicals turned the waste into manure, showing biogas and manure are produced by the same air-loving bacteria

46. Order these four EVENTS in the life of a sewage drop by when they happen: (i) a flammable gas is tapped from a sealed dome, (ii) heavy sand sinks out in a slowed flow, (iii) the remaining dissolved organic matter is eaten by air-using bacteria, (iv) a torn plastic sheet is stopped at the entry bars.

- (A) ii, then iv, then i, then iii (B) iv, then ii, then i, then iii
- (C) i, then iv, then ii, then iii (D) iv, then ii, then iii, then i

47. A claim: 'Because treated effluent can be used to recharge groundwater, and groundwater is what wells draw for drinking, the effluent is therefore already drinking water.' Where does the logic fail?

- (A) The logic is sound, so plants should pipe their effluent straight to drinking taps
- (B) The logic fails only because groundwater is never used for drinking anywhere
- (C) Recharge lets the not-yet-potable effluent soak through soil; it is the natural ground filtration over time, not the effluent itself, that the system relies on — the effluent leaving the plant is still not certified drinking water
- (D) The logic fails because the effluent is actually dirtier than raw sewage and must never touch the ground

48. A worker must identify which tank is which using only two clues: Tank 1 shows a sludge layer below and an oily film above with NO bubbling; Tank 2 shows vigorous bubbling and clearing water with NO settled-sludge-and-scum layering. Which identification and reasoning is correct?

- (A) Tank 1 is the aeration tank because the oily film proves bacteria are active, and Tank 2 is the settling tank
- (B) Tank 1 is the sludge digester and Tank 2 is the grit chamber, since one is layered and one is stirred

(C) Both tanks are aeration tanks, since both clear the water in the end

(D) Tank 1 is the settling tank (sludge sinks, scum floats, no air needed) and Tank 2 is the aeration tank (air bubbled to grow bacteria that clear the water)

49. A municipality with limited budget must rank these four problems by how directly each spreads waterborne disease, to fix the worst first: (i) sewage leaking into a drinking well, (ii) a noisy aeration blower, (iii) a slightly rusty bar-screen frame, (iv) a faded sign at the plant gate. Which is the clear top priority and why?

(A) Problem (ii), the noisy blower, because noise is the chapter's main cause of waterborne disease

(B) Problem (i), sewage leaking into the drinking well, because mixing sewage with drinking water is the direct route by which waterborne diseases like cholera and typhoid spread

(C) Problem (iii), the rusty frame, because rust on a screen is what poisons the drinking water

(D) Problem (iv), the faded sign, because unreadable signs are how germs enter the water supply

50. A student writes: 'Aerobic bacteria do the work inside the sealed digester, and anaerobic bacteria do the work inside the aerated tank.' Exactly which part(s) must be changed to make it correct?

(A) Only the digester half needs changing; the aerated-tank half is already correct

(B) Only the aerated-tank half needs changing; the digester half is already correct

(C) Both labels must be swapped: anaerobic bacteria work inside the air-free sealed digester, and aerobic bacteria work inside the aerated tank

(D) Nothing needs changing; the statement is already correct as written

51. A plant takes inlet (raw sewage) and outlet (treated) samples. The lab reports the outlet has much less organic matter and far fewer solids than the inlet. A reporter writes the headline 'Plant proves treated water now safe to drink.' Why is the headline an overreach?

(A) Lower organic matter and solids show the water is much cleaner, but that does not certify it free of all germs, so 'safe to drink' is not justified by those results

(B) The headline is fully justified, because fewer solids automatically means zero germs

(C) The headline is wrong because the outlet sample is actually dirtier than the inlet

(D) The headline is wrong because treatment changes nothing, so the two samples are identical

52. A claim: 'A septic tank is just a household-sized sewage treatment plant, so its water can be drunk and it never needs servicing.' Identify the two errors.

(A) Only the drinking claim is an error; it truly never needs servicing because solids vanish on their own

(B) Only the servicing claim is an error; its water is genuinely safe to drink straight away

(C) Both errors: a septic tank only partly treats waste on site so its output is not drinking water, and its settled sludge accumulates and must be pumped out periodically

(D) Neither is an error; a septic tank does make drinking water and never accumulates sludge

53. A plant manager says: 'Recycling activated sludge back into the aeration tank is wasteful — we are just putting dirt back in.' Why is the recycling actually deliberate and useful?

(A) The recycled activated sludge is the population of working aerobic bacteria, and returning it keeps a strong, active bacterial community cleaning the incoming dissolved organic matter

(B) Recycling puts inert grit back so it can be re-settled, which is the real reason for the loop

(C) Recycling returns biogas to the tank so the water can be made flammable

(D) Recycling pours raw untreated sewage back to dilute the clean water and slow the bacteria down

54. A worker must decide, for safety, which ONE removed material may be spread on farmland as manure and which must instead be sent to disposal. The candidates are dried digested sludge, screened plastic-and-rag debris, and grit. Which choice is right?

(A) Screened plastic-and-rag debris makes the best manure, while the dried digested sludge must be thrown away

(B) Dried digested sludge may be used as manure, while screened plastic-and-rag debris and grit must be disposed of, not spread on fields

(C) Grit is the proper manure for fields, while the dried digested sludge must be disposed of

(D) All three may be spread on fields as manure since all came out of the sewage

55. A claim about disinfection: 'Chlorination at the end is unnecessary because the aeration tank's bacteria have already eaten every germ.' Why does the chapter still call for the disinfection step?

(A) Disinfection is needed only to add nutrients, since the bacteria leave the water too pure

(B) Disinfection is needed only to settle out the heavy grit the bacteria stirred up

(C) Disinfection is genuinely unnecessary, because aerobic bacteria do kill every germ

(D) Aeration's bacteria consume dissolved organic matter but do not reliably destroy the disease-causing microbes, so a separate disinfection step is needed to kill the germs that survive

56. A drop of dishwater is traced through the system from the house to the environment. Put these stages in the true order: (i) treated water recharges groundwater, (ii) the drop

enters the underground sewer, (iii) aerobic bacteria clean dissolved organics, (iv) bulky debris and grit are removed by physical units.

- (A) ii, then iii, then iv, then i
- (B) ii, then iv, then iii, then i
- (C) iv, then ii, then iii, then i
- (D) ii, then iv, then i, then iii

57. A factory argues, 'Our pipe puts chemicals only into the sewer, never directly into the river, so the river is protected.' Why does the chapter reject this defence?

- (A) Chemicals in the sewer can kill the plant's treatment bacteria and may pass through to the river anyway, so dosing the sewer does not protect the river and also harms the treatment process
- (B) The defence holds, because anything entering a sewer is fully neutralised before reaching any river
- (C) The defence holds, because the bar screen removes all dissolved chemicals before treatment
- (D) The defence fails only because sewers are not connected to treatment plants at all

58. A planner compares three on-site sanitation choices for places without sewers — a septic tank, a vermi-processing toilet, and a chemical toilet — and asks which one yields a useful product as part of its normal working. Which statement is correct?

- (A) The chemical toilet yields drinking water as its useful product, while the others yield nothing
- (B) The septic tank yields cooking gas piped to the kitchen as its normal product, while the others yield nothing
- (C) All three yield biogas as their main useful product in identical amounts
- (D) The vermi-processing toilet yields useful manure from the worms' action, while the chemical toilet mainly breaks down and deodorises waste and the septic tank chiefly stores and partly treats waste needing periodic desludging

59. A teacher asks for the ONE statement that correctly separates a PHYSICAL-mechanical removal, a BIOLOGICAL-aerobic removal, and a BIOLOGICAL-anaerobic process, each with its right material.

- (A) Settling tank physically settles suspended sludge; aeration tank aerobically lets bacteria eat dissolved organics; digester anaerobically decomposes sludge to biogas
- (B) Settling tank aerobically grows bacteria; aeration tank physically settles sludge; digester physically screens debris
- (C) Settling tank anaerobically makes biogas; aeration tank physically removes grit; digester aerobically eats dissolved organics
- (D) Settling tank physically removes dissolved salts; aeration tank anaerobically makes manure; digester aerobically settles sludge

60. A student concludes, 'Since the chapter says better sanitation prevents disease, doctors and medicines are pointless and we should rely on sanitation alone.' Why is this the wrong lesson to draw?

- (A) The chapter favours prevention because keeping sewage out of drinking water stops waterborne disease before it starts, which is better than only treating the sick — but it does not claim treatment of illness is pointless
- (B) It is wrong because the chapter actually says doctors can cure no waterborne disease at all
- (C) It is wrong because clean drains generate biogas that hospitals burn instead of giving medicine
- (D) It is right; the chapter does say sanitation completely replaces all medical care

Solutions — Science (Class 7), Chapter 18:

Wastewater Story — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	C	31	B	41	B	51	A
2	B	12	D	22	A	32	B	42	B	52	C
3	B	13	A	23	D	33	C	43	B	53	A
4	D	14	A	24	A	34	D	44	C	54	B
5	C	15	D	25	A	35	C	45	B	55	D
6	D	16	B	26	C	36	A	46	D	56	B
7	B	17	D	27	B	37	C	47	C	57	A
8	D	18	B	28	A	38	C	48	D	58	D
9	C	19	D	29	D	39	A	49	B	59	A
10	D	20	C	30	A	40	C	50	C	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) The residue left after boiling a clear filtrate is made of substances that were dissolved and survive evaporation — dissolved inorganic salts like nitrates and phosphates, which the chapter classes as inorganic impurities. Organic matter (food bits, faeces, oil) is carried as solids or floating film, here in fractions (i) and (ii), so it is misplaced to call the salt residue organic. The residue is not dried-out microbes — pathogens are the living fraction (iv), not a salt cake. Dissolving in water does not convert a salt into an organic compound; classification depends on the substance, not on whether it dissolves. And not everything in sewage is organic — the very point of the sorting is that dissolved inorganics form a separate class.

2. (B) Pathogens are living microbes that enter chiefly with human and medical waste, so a clinic and households dominate that class. Dissolved inorganic salts (nitrates, phosphates, metal salts) come largely from industrial effluent like a dye-mill and from fertiliser-laden field runoff, so those dominate the dissolved-inorganic class. Fields contribute nutrients and runoff, not the heaviest pathogen load. A dye-mill releases

chemicals and dissolved salts, not the main microbe load, and a clinic certainly does not add only clean rainwater. The sources are not interchangeable; the whole reason sewage is mixed is that different sources add different dominant contaminant classes.

3. (B) Large objects are caught first by the bar screen (the rag), then heavy sand and grit settle in the grit chamber, then in the settling tank the sludge sinks while the oily scum is skimmed off the top, then the clarified water is aerated so aerobic bacteria consume dissolved organic matter, and finally the settled sludge is sent to the air-free digester where anaerobic bacteria release biogas. Grit removal never precedes screening, because screens protect the works from the bulky debris first. Aeration of dissolved organics happens after, not before, the scum-and-sludge separation in the settling tank. And settling/skimming come after grit removal, not before it.

4. (D) Each unit has a distinct job. The bar screen stops bulky floating objects like a sack. The grit-and-sand chamber lets heavy mineral particles sink by slowing the flow. The settling (sedimentation) tank captures the fine suspended solids that slipped past the grit chamber, settling them as sludge. The aeration tank is where aerobic bacteria consume the dissolved organic matter that no physical step can remove. The grit chamber cannot catch a sack, nor does the bar screen catch grit. The settling tank, not the aeration tank, removes fine suspended mud, and dissolved organic matter is consumed by aeration, not settled out or digested in the sludge digester.

5. (C) Bubbling air means oxygen-needing aerobic bacteria, and their job is to consume dissolved organic matter so the water clears — that is Tank P. Sealing from air means oxygen-free anaerobic bacteria, and their action on sludge releases biogas — that is Tank Q. The trainee swapped both: he called the aerated cleaning tank anaerobic and biogas-making, and the sealed gas-making tank aerobic and cleaning, so both claims are wrong. It is not just one error; both halves are reversed. And the two tanks do not use the same bacteria — they deliberately use opposite types under opposite air conditions.

6. (D) Biogas comes from anaerobic bacteria decomposing organic sludge, so only the organic settled solids feed the digester usefully. Grit is inert mineral sand that the bacteria cannot ferment into gas, and it is taken out early precisely so abrasive sand neither wears equipment nor contaminates the sludge. So feeding grit to the digester is pointless. The digester acts on settled organic sludge, not on the skimmed floating scum. And settling does not transform sand into organic matter — what settles in the grit chamber stays mineral grit.

7. (B) Treatment removes most solids and organic matter, so the effluent looks much cleaner than raw sewage, but the chapter is explicit that cleaner is not the same as potable — the treated water may still carry germs and is not declared safe to drink. Looking clear does not certify it as drinking water. Treatment makes the water cleaner, never dirtier than the raw inlet. And added clearness does not mean added nutrients making it safe; extra dissolved nutrients would be a pollution problem, not a sign of drinkability.

8. (D) Bulky floating debris like plastic and branches is the bar screen's job, and heavy washed-in road sand is the grit chamber's job — those two units bear the brunt of a storm surge, so they are checked first. The aeration tank grows bacteria and is not built to trap plastics, and the digester decomposes organic sludge, not sand. The settling tank captures fine suspended solids, not bulky plastics, and the aeration tank does not remove grit. Matching each load to the unit physically designed for it points to the screen and the grit chamber.

9. (C) Grit removal sits early precisely to stop abrasive mineral sand from wearing moving equipment and from mixing into the organic sludge handled later, which would degrade the sludge and the products derived from it. Sand is inert, not bacterial food, and it does not make effluent drinkable. Mineral sand does not dissolve into nitrate, and bacteria cannot turn sand into biogas. The bar screen is upstream of the grit chamber and has coarse bars that pass sand, so it cannot rescue the situation downstream — the sand reaches the later units and does the damage.

10. (D) A floor scraper moving settled solids plus a surface skimmer taking off a greasy film, with no air being bubbled, describes the settling tank, where the dense bottom material is sludge and the floating film is scum. The aeration tank would show vigorous bubbling and is not where scum-and-sludge layers are skimmed and scraped this way; biogas is a gas, not a surface film here. Bottom and top must not be swapped — sludge sinks and scum floats. And the sealed digester has no open scraping-and-skimming arms exposed like this, nor does it hold grit and 'manure' as its two layers.

11. (C) The aeration tank is the only step where aerobic bacteria consume the dissolved organic matter that physical steps cannot remove, so skipping it leaves that dissolved organic load in the water. Plastics are still caught because the bar screen is untouched. Grit is still removed because the grit chamber remains. The bar screen still works — none of the screening, grit removal or settling is what was deleted. The unique loss from dropping aeration is the unremoved dissolved organic matter.

12. (D) The chlorination step is a final disinfection to destroy harmful microbes that physical and biological treatment did not remove; it targets pathogens, not dissolved salts. Nitrates and phosphates are a dissolved inorganic load that chlorination is not designed to take out. Settling heavy grit is the grit chamber's job, done early, not chlorination's. Chlorination does not make biogas — that is the anaerobic digester. And chlorine kills germs rather than washing away every dissolved salt, so the original claim is mistaken.

13. (A) After settling, the sludge (the solids) is decomposed in the sealed, air-free digester by anaerobic bacteria, producing biogas and a digested sludge later dried as manure, while the clarified water (the liquid) is sent to the aeration tank where aerobic bacteria clean out its dissolved organics. The two streams must not be swapped: water is not digested for biogas, and sludge is not aerated. The streams are handled in different units, not together

in one digester. And the clarified water is not turned into manure, nor is sludge dumped back into the inlet.

14. (A) A village biogas unit works by anaerobic (air-free) decomposition that releases biogas, exactly like the sealed sludge digester at a treatment plant. The aeration tank is the opposite — it uses air and does not produce biogas. The grit chamber is a physical settling unit using no sunlight and yielding grit, not manure or gas. The bar screen merely sieves large debris and produces no flammable gas. Only the anaerobic digester matches both the air-free condition and the biogas product.

15. (D) Faeces is decaying organic matter; dissolved nitrate is a dissolved inorganic salt; intestinal bacteria are living disease-causing microbes — each fits one box cleanly. Faeces is organic, not a dissolved salt; nitrate is an inorganic salt, not a living thing; and bacteria are living pathogens, not dissolved inorganics, so the second option scrambles all three. The third option ignores that faeces and bacteria are not dissolved salts at all. The chapter's three contaminant classes map one-to-one onto these three examples.

16. (B) Dissolved organic matter passes straight through screening, grit removal and settling, all of which act on solids; only the aeration tank's aerobic bacteria consume it. So a high dissolved-organic outflow, with the physical units normal, points to failing aeration. The bar screen strains bulky objects, not dissolved matter. The grit chamber settles mineral sand and does nothing to dissolved organics. The settling tank removes suspended sludge, not dissolved organic matter. By elimination, aeration is the culprit.

17. (D) A septic tank is the standard low-cost on-site option where no sewer exists; solids settle and partly decompose inside it, but sludge builds up and must be pumped out periodically and disposed of or treated. A full municipal plant per house is neither cheap nor maintenance-free. A bar screen only sieves debris and cannot make drinking water. A household grit chamber would just collect sand, not generate cooking gas. Only the septic tank fits the on-site, low-cost description, and its upkeep is regular desludging.

18. (B) Biogas is a gas produced when anaerobic bacteria decompose sludge in the sealed digester; scum is the floating oily and greasy material skimmed off the open settling tank's surface — they differ in physical state, in origin process and in use, so they are not the same substance. The reason offered in the first distractor inverts the facts (biogas is not bottom sludge; scum is not a gas). Scum is indeed a floating layer, but biogas is not, so they are not both 'the floating oily layer.' And only biogas is a flammable gas from a sealed dome; scum is not, so they are not both gases.

19. (D) Treatment leans on living bacteria in the aeration tank and the digester; toxic chemicals can poison and kill those microbes, so the biological stages suffer most directly. The chemicals do not improve screening — the screen is a passive physical sieve unaffected in the way claimed. Dissolving grit is not the harm the chapter highlights, and grit removal

is physical, not biological. The chemicals certainly do not transform sewage into safe manure; that fantasy ignores the real danger to the bacteria the plant relies on.

20. (C) A sealed, air-free dome releasing flammable gas is the anaerobic sludge digester producing biogas, while an open tank with vigorous bubbling that clears the water is the aerobic aeration tank, whose bacteria yield cleaned water and settle as activated sludge. The labels must not be swapped — the aeration tank is the bubbling one, not the sealed dome. A settling tank does not bubble and a grit chamber does not aerate, so the third pairing is wrong. And the two units are not both digesters; one is aerobic aeration, the other anaerobic digestion.

21. (C) The trouble is physical, not biological: solid scraps and tea leaves accumulate and clog the pipes, blocking the flow long before any treatment bacteria could act on them, so 'they are organic' does not save the situation. The scraps do not dissolve instantly nor improve water quality. There are no treatment bacteria living in household taps to eat them. And the practice is not harmless — the chapter warns against putting solids down the drain precisely because of the clogging.

22. (A) Waterborne diseases like cholera and typhoid spread when sewage contaminates drinking water, exactly what leaking sewers into wells would cause, so the cure is better sanitation that separates sewage from drinking-water sources. Household biogas does not cause these outbreaks. Recharging clean-enough treated water into the ground is a beneficial reuse, not the cause of disease. And removing bar screens would worsen, not fix, the plant — it has nothing to do with the well contamination. The chain is sewage-into-drinking-water, fixed by sanitation.

23. (D) The settled mass of aerobic bacteria that has cleaned the organic matter is called activated sludge, and a large portion is recycled back into the aeration tank so the established bacterial population keeps the cleaning going. It is not grit — grit is inert mineral sand removed earlier. It is not scum — scum is the floating oily film, not the settled bacterial mass. And it is certainly not 'raw sewage' to be dumped back at the inlet; recycling activated sludge is a controlled step that maintains the active bacteria.

24. (A) A vermi-processing toilet relies on earthworms to break human waste down into useful manure right where it is produced, avoiding the discharge of untreated waste — that on-site, no-pollution conversion is what makes it environment-friendly. Flushing faster and further into a river spreads pollution, the opposite of the benefit. It does not burn waste or make biogas; it uses worms, not combustion. And it is not an air-pumped aerobic mini-plant; the defining feature is the worms producing manure.

25. (A) The settling tank is where fine suspended solids that escaped the grit chamber are removed as sludge; omit it and those fine solids pass into the aeration tank, overloading the biological step rather than being settled out first. Dissolved organic matter can still be removed — aeration is retained. Plastics are still caught by the untouched bar screen. Grit

is still removed by the untouched grit chamber. The specific loss from dropping settling is that the fine suspended solids are no longer taken out before aeration.

26. (C) The settling tank settles out raw sludge — the mixed organic solids that sank — whereas the aeration tank's settled solids are activated sludge, the mass of aerobic bacteria that did the cleaning. The two are not the same grit; mineral grit was removed earlier in its own chamber. Biogas is a gas made in the sealed digester, not a bottom solid in either tank. And scum is a floating film, not a bottom solid, so neither bottom holds scum. The key contrast is raw settled sludge versus living activated sludge.

27. (B) The chapter presents groundwater recharge as a sensible reuse: the treated water, while not certified for drinking, is clean enough to soak safely into the ground and replenish supplies. It is not pure drinking water — that is the whole reason it is recharged rather than piped to taps. It is far cleaner than raw sewage, not dirtier, so 'hiding' dirtier water is wrong. And soaking water into soil does not generate biogas — biogas comes from anaerobic digestion of sludge, not from infiltration.

28. (A) The bar screen takes out bulky floating objects and the grit chamber takes out heavy sand, but plenty remains: fine suspended solids (later settled as sludge), the floating scum, and the dissolved and microscopic loads that need the settling tank, aeration and disinfection. So the claim that every solid is gone is false. The bar screen does not add solids back. It is not correct that the two units remove every solid. And the grit chamber removes mineral grit, not dissolved salts, so that reason is also wrong.

29. (D) Large floating debris leaves first at the bar screen, then heavy grit leaves at the grit chamber, then settled sludge is drawn from the settling-tank floor, and last the dissolved organic matter is consumed by bacteria in the aeration stage. Sludge cannot leave before grit, because grit removal precedes settling. Grit does not precede the screening of bulky debris. And dissolved organic matter is removed last, not first, so the order beginning with iv is reversed.

30. (A) Nitrates and phosphates act as nutrients that over-feed water plants; their runaway growth disturbs the balance of the water body and harms the other organisms, matching the overgrowth-and-fish-death picture. Such nutrient pollution does not make water drinkable — and 'too pure' is not what kills fish here. Dissolved nutrients do not harden into sand. And ponds do not turn nutrients into biogas; biogas comes from anaerobic digestion of sludge, not from nutrients in open water.

31. (B) A chemical toilet adds chemicals to decompose the waste and reduce its smell and germs right where it is, which is exactly what suits a remote site with no sewer or treatment plant. It does not produce drinking water. It does not rely on a powered aerobic system — that is the point of using chemicals instead of an aeration tank that needs infrastructure. And it is not a biogas collector piped to a kitchen; its job is on-site breakdown and odour/germ control.

32. (B) The two valuable outputs are biogas, the flammable fuel from anaerobic digestion of sludge, and the dried digested sludge used as manure — both from the digester. Screened plastics and inert grit are wastes to dispose of, not a building material and fertiliser. Scum is a greasy skimmed film, not a clean cooking fuel, and grit is inert sand, not a soil nutrient. Raw untreated sludge must never be tipped into a river, and chlorine is a disinfectant, not a drink additive. Only biogas and manure are the genuine recovered products.

33. (C) Plain rinse water carries neither pipe-clogging solids nor bacteria-killing toxins, so it is the safe act. Paint thinner is a solvent that harms the treatment bacteria, not a helper. Hot frying oil cools and coats the pipes, clogging them, so heat does not save it. Expired medicines are chemicals that can poison the plant's bacteria rather than being harmlessly eaten. Each of the other three either clogs pipes or threatens the bacteria, leaving the rinse water as the least harmful.

34. (D) When drains are cleared they do not overflow, so disease-laden waste is kept away from living spaces and is less able to contaminate the water people use — that is how the routine cuts disease. The work reduces, not raises, the microbes reaching people; it does not 'train immunity' by spreading germs. The crew is not dosing household sinks with chemicals. And clearing drains has nothing to do with burning kitchen biogas. The real mechanism is keeping waste away from people and their water.

35. (C) The bar screen is a passive physical sieve that needs no bacteria, and the aeration tank is the biological stage where aerobic bacteria consume dissolved organics — both halves are true. The grit chamber removes mineral grit, not dissolved nitrates, and the digester decomposes sludge, not plastics. The settling tank does not make biogas, and the aeration tank does not make grit. And it is chlorination, not the bar screen, that kills microbes, while the bar screen, not chlorination, catches large debris — so only the first statement is fully correct.

36. (A) The digester is sealed precisely to exclude air, creating the oxygen-free conditions that anaerobic bacteria need to decompose sludge and give off biogas — so sealing favours anaerobic, not aerobic, action. A sealed tank does not 'trap fresh air' to feed aerobes; it shuts air out. The digester and aeration tank do not use identical bacteria — they use opposite types. And a sealed digester does not let sunlight in to dry sludge; it runs anaerobic bacterial digestion. The student's premise about sealed places trapping usable air is the error.

37. (C) Open defecation leaves waste where rain can wash it and its pathogens into nearby water; here the uphill site drains into the stream the village drinks, so the germs contaminate drinking water and cause waterborne disease. Exposed waste does not simply become harmless biogas in the open. It does not add 'clean' water — it pollutes the supply. And there are no helpful bacteria in household taps to be fed; the chain is rain-borne contamination of the drinking source.

38. (C) Aerobic bacteria need the oxygen the blower supplies; without it they are starved, so they break down the dissolved organic matter poorly and their population falls. They do not thrive better without air — air is what they require. The aeration tank does not switch to making biogas; biogas comes from the sealed anaerobic digester. And gravity cannot filter out dissolved organic matter — only the bacteria can consume it, which is exactly why losing the blower is harmful.

39. (A) The bar screen removes bulky debris by physical sieving, with no organisms involved. The aeration tank is biological — its bubbling supplies oxygen to living aerobic bacteria that do the cleaning, so it is not purely mechanical. The sludge digester is also biological — sealing it enables anaerobic bacteria, not removes organisms. Recycling activated sludge returns living bacteria to keep the biology going, so it is biological too. Only the bar screen is purely physical.

40. (C) Activated sludge is the living aerobic-bacterial mass that settles in the aeration tank and is largely recycled there, while manure is the dried, stabilised residue of sludge after anaerobic digestion in the sealed digester — they come from different tanks and processes, so they are not the same material. They are not both dried solids from one tank. Activated sludge is not a gas and manure is not grit. And neither is the floating scum, which is a separate skimmed layer. The correct contrast is living aeration-tank bacteria versus digested-and-dried sludge residue.

41. (B) Scum is specifically the floating oily and greasy film skimmed from the settling tank's surface; the digester's product is biogas, a flammable gas formed by anaerobic decomposition and drawn from the sealed dome — so only the film is scum. 'Anything that rises is scum' is false; a gas is not scum. The film is not biogas and the dome gas is not sludge, so swapping them is wrong. And they are not both biogas — one is an oily skimmed layer, the other a gas. They differ in state and origin.

42. (B) Physical and biological steps reduce solids and organic load but do not reliably kill microbes; the chlorination/disinfection step is the one meant to destroy the remaining harmful germs, so a microbe spike with everything else normal points to failed disinfection. Removing grit does not kill microbes. Straining bulky debris at the screen does not remove microscopic organisms. Aeration grows bacteria to eat organics; it is not the germ-killing step. Only disinfection failure specifically explains the surge in microbes.

43. (B) The chapter splits the work by contaminant class: physical screening and settling take out bulky and suspended solids, aeration's aerobic bacteria consume dissolved organic matter, and disinfection handles the surviving microbes. The bar screen cannot remove dissolved matter or microbes — only bulky objects. Dissolved organic matter is not screened, and bulky solids are not eaten by bacteria — that reverses the roles. Microbes are not settled out as sludge, and the bar screen does not catch dissolved salts. Only the first statement maps each class to the right stage.

44. (C) Order matters because abrasive grit reaching the aeration tank would damage its equipment and could smother the bacteria, so the grit chamber must act first. Sand is inert and is not bacterial food, so the bacteria do not need it. The grit chamber, not the aeration tank, is the unit built to remove grit. And grit does not need aerating to settle — it sinks because it is heavy mineral matter. The sequence protects the biological stage, which is why grit removal precedes aeration.

45. (B) Biogas is produced by living anaerobic bacteria, so harsh bleach and acid that poison them stop the gas — revealing the process depends on microbes that toxins can kill, just as toxic effluent harms a treatment plant's bacteria. Biogas is not a purely chemical reaction needing no organisms. Adding oxygen would hinder, not help, anaerobic bacteria. And the chemicals did not helpfully make manure; the point is that the bacteria were destroyed. The collapse demonstrates the biological, microbe-dependent nature of biogas generation.

46. (D) The plastic sheet is stopped first at the bar screen, then heavy sand sinks out in the slowed grit-chamber flow, then the clarified water is aerated so air-using bacteria eat the dissolved organics, and finally the settled sludge sent onward is digested anaerobically, tapping biogas from the sealed dome. Sand removal does not precede the screening of bulky debris. Biogas tapping comes after aeration of the water, since it is the downstream sludge that is digested. And biogas tapping cannot be first — the sludge that makes it has not yet been produced and settled.

47. (C) The effluent is recharged because it is clean enough to soak safely into the ground, but it is not itself certified drinking water — the system depends on the water re-entering the natural ground store, not on the plant declaring the effluent potable. So 'it goes underground, therefore it is drinking water' skips the point that it was never certified safe to drink. Plants do not pipe effluent straight to taps for that very reason. Groundwater is in fact used for drinking via wells, so that denial is false. And the effluent is far cleaner than raw sewage, not dirtier, so it may safely be recharged.

48. (D) A sludge-and-scum layered tank with no bubbling is the settling tank, where solids sink as sludge and the oily film floats as scum; a vigorously bubbling tank that clears the water is the aeration tank, where air feeds the cleaning bacteria. The oily film is scum, not proof of active bacteria, so Tank 1 is not aeration. The digester is sealed (not openly layered) and the grit chamber settles sand (not stirred to clear water), so that pairing is wrong. And the two tanks are not both aeration tanks — only one is bubbled. The bubbling-versus-layering clue distinguishes them.

49. (B) Waterborne disease spreads when sewage contaminates drinking water, so a sewage leak into a drinking well is the most direct disease pathway and the clear top priority. A noisy blower is a maintenance nuisance, not a disease route — noise does not spread cholera. Surface rust on a screen frame does not poison drinking water. A faded

gate sign has no bearing on how germs reach the water. Only the sewage-into-drinking-water leak fits the chapter's mechanism for waterborne disease.

50. (C) The sealed digester is air-free, so it hosts anaerobic bacteria, while the aerated tank is full of air, so it hosts aerobic bacteria — the student has both backwards, so both halves must be swapped. It is not just the digester half — the aerated-tank half is also wrong. It is not just the aerated-tank half — the digester half is wrong too. And the statement is certainly not already correct, since it reverses both roles.

51. (A) Reduced organic matter and solids prove the water is much cleaner, but they say nothing about whether all germs are gone, so declaring it drinkable overreaches — cleaner is not certified potable. Fewer solids does not automatically mean zero germs, which is exactly the gap. The outlet is cleaner, not dirtier, than the inlet. And treatment plainly does change the water — the samples are not identical — so that denial is also wrong. The headline confuses 'cleaner' with 'safe to drink.'

52. (C) A septic tank gives only partial on-site treatment, so its effluent is not drinking water, and sludge builds up inside it and must be pumped out periodically — so both parts of the boast are wrong. Solids do not vanish on their own; they accumulate and need desludging, so the 'never needs servicing' half is false. The water is not safe to drink straight away, so the drinking half is also false. Neither claim survives, so saying neither is an error is incorrect.

53. (A) Activated sludge is the mass of aerobic bacteria that did the cleaning; recycling it seeds the aeration tank with a ready, active bacterial community so incoming dissolved organics are consumed efficiently — a deliberate, useful step, not putting 'dirt' back. It is not grit being re-settled; grit was removed earlier and is not what is recycled. It is not biogas, which is a gas from the digester, not something fed to the aeration tank. And it is not raw sewage diluting clean water; it is the living bacteria that drive the treatment.

54. (B) Only the dried digested sludge, the stabilised residue from anaerobic digestion, is used as manure; the screened plastics and rags and the inert grit are non-nutrient wastes that must be disposed of, not spread on land. Plastic-and-rag debris is not manure — it is bulky inorganic waste. Grit is inert mineral sand, not a soil nutrient. And not everything that came out of the sewage is fit for fields; the bar-screen debris and grit are wastes, while the digested sludge is the useful product.

55. (D) The aeration bacteria feed on dissolved organic matter; they do not reliably wipe out the pathogens, so a dedicated disinfection step is added to kill the surviving germs before release. Disinfection is for germ-killing, not for adding nutrients. It is not for settling grit, which is removed early in the grit chamber. And it is not unnecessary — the aerobic bacteria do not eliminate all germs, which is the whole reason disinfection exists. The claim misjudges what the aeration bacteria actually do.

56. (B) Dishwater first enters the underground sewer, then at the plant the physical units (bar screen and grit chamber) take out bulky debris and grit, then aeration's aerobic bacteria clean the dissolved organics, and finally the treated water can recharge groundwater. Aeration cannot precede the physical removal of debris and grit, which protect the works. The drop reaches the sewer before any treatment, so the sewer step comes before the physical units, not after. And groundwater recharge is the last stage, after the water has been cleaned, not before aeration.

57. (A) Putting chemicals into the sewer is not safe: they can poison the bacteria the plant needs, and chemicals not handled by treatment can still pass on to the river, so the river is not protected and the process is harmed too. Sewers do not fully neutralise everything before a river — that is wishful. The bar screen sieves bulky solids and cannot remove dissolved chemicals. And sewers do connect to treatment plants, so that reasoning is false. The defence ignores the damage to the bacteria and the carry-through to the river.

58. (D) Among these on-site options, the vermi-processing toilet routinely produces useful manure as earthworms convert the waste; the chemical toilet's role is to break down waste and cut odour and germs where no sewer exists; and a septic tank stores and partly treats waste, needing periodic pumping of its sludge. The chemical toilet does not make drinking water. A septic tank's normal output is not piped cooking gas. And the three do not all make equal biogas — that is not the defining product of any of them in this comparison. Only the vermi toilet's manure fits 'useful product of normal working.'

59. (A) The settling tank is physical — suspended solids settle out as sludge by gravity. The aeration tank is biological-aerobic — air-fed bacteria consume dissolved organics. The digester is biological-anaerobic — air-free bacteria decompose sludge to biogas. The other options scramble these: the settling tank does not grow bacteria or make biogas, the aeration tank does not settle sludge or remove grit, and the digester does not screen debris or settle sludge. Only the first statement pairs each stage with its true nature and material.

60. (A) The chapter's point is that prevention — keeping sewage away from drinking water — stops disease before it starts and is better than relying only on curing the sick; it does not declare medical treatment pointless. It does not claim doctors can cure no waterborne disease. Clean drains have nothing to do with hospitals burning biogas instead of treating patients. And it does not say sanitation completely replaces medical care; the lesson is that prevention is preferable, not that cure is worthless. Over-reading 'prevention is better' into 'cure is pointless' is the error.